

Gemedi: Discharge Medical Note Generator

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I. ABSTRACT

Access to real clinical text containing authentic protected health information (PHI) is extremely limited. However, these data are essential for developing and evaluating medical language models. At first glance, large language models seemed to be a natural solution: with a simple prompt, they can generate discharge notes that look super realistic. However, we repeatedly encountered a frustrating trade-off in initial testing. Prompts that preserve the natural structure and flow of a medical note often produce medically illogical content. While prompts designed to enforce strict medical correctness tend to break the document’s realism, the outputting text can feel to be rigid or unstructured. This balance between linguistic realism and medical reasoning makes it difficult to generate synthetic medical notes that are both believable and clinically coherent.

To address this problem, we introduce **Gemedi**, a discharge medical note generation framework built around a dual-discriminator feedback loop. Rather than relying on a single or a overly complex prompt, Gemedi separates evaluation into two complementary roles. One discriminator focuses on structural realism and common formatting or field-level errors, which is trained using deliberately flawed synthetic data with annotated reasoning mistakes. The second discriminator evaluates deeper medical consistency and semantic logic using the Google Gemini API. The generator itself is fine-tuned on high-quality synthetic notes and then iteratively refines its outputs by incorporating feedback from both critics through prompt updates. This separation allows each component to focus on what it does best, which allows the generator to improve realism and medical reasoning simultaneously. Our results show that Gemedi produces synthetic discharge notes that maintain high linguistic fidelity while achieving comparable or superior medical reasoning performance compared to baselines, suggesting a practical path towards high-quality synthetic clinical text generation.

 **GitHub Repository:** <https://github.com/yumcui/Gemedi>

II. INTRODUCTION

Today, High-quality clinical text is essential for developing and evaluating medical language models. However, accessing to real medical notes containing authentic protected health information (PHI) is severely restricted. Privacy regulations such as HIPAA make real clinical data difficult to obtain, even

widely used public datasets like MIMIC-IV [2] are fully de-identified. Many important details like patient names, dates, and unit numbers are removed, which are the key factors for training medical language models. As a result, synthetic medical note generation has become an increasingly important tool for medical NLP research.

LLMs generated medical notes seemed to be the solution for synthetic data, but in practice we observe a persistent trade-off. Applying simple prompts with one-shot strategy often can preserve document structure and linguistic fluency. However, it introduces medically illogical or inconsistent content. In contrast, prompts that enforce stricter medical constraints frequently degrade realism, making text feel artificial and unstructured.

We argue that this challenge arises from coupling two distinct objectives within a single generation process. Structural and linguistic realism concerns formatting, section layout, and consistency at words-level. In comparison, medical reasoning requires clinically valid relationships across the note. Existing approaches typically attempt to address both objectives using a single prompt, which leads to unstable trade-offs and limited controllability.

We introduce **Gemedi**, a discharge medical note generation framework built around a dual-discriminator feedback loop. The generator is fine-tuned on high-quality synthetic notes constructed from de-identified clinical text with injected fake PHI. Evaluation of generated notes is separated into two complementary roles. It includes a realism-focused discriminator trained on deliberately flawed synthetic notes with annotated reasoning errors, and a medical reasoning discriminator implemented using the Google Gemini API. Feedback from both discriminators is used to iteratively refine generation through prompt updates. This approach introduces improvements in realism and medical reasoning to proceed independently. Our experiments show that Gemedi produces synthetic discharge notes with high linguistic while maintaining comparable or superior medical reasoning performance relative to standard prompting baselines and generator-only framework.

A. Motivation

Our motivation comes from our observation of a paradox that the current medical AI community faces. The most valuable training data is the least accessible due to privacy concerns, while available sanitized datasets fail to capture the complexity of real clinical documentation. We think it’s important to the clinical and medical field to develop stronger

practical clinical NLP systems, including medical chatbots and automated documentation tools. By creating synthetic notes that mirror real PHI patterns without compromising patient privacy, we can enable more robust training of systems that will eventually support clinicians in real healthcare environments.

B. Goals

When starting this project, we defined our goals in three stages. Our base goal was to ensure that introducing a fine-tuned generator and discriminator-based feedback would not degrade performance relative to standard LLM-based generation, which it aims to achieve comparable levels of linguistic realism and medical reasoning. Our target goal was to surpass non-fine-tuned LLM baselines by simultaneously achieving high structural realism and reasonable medical reasoning. This goal is built on improving one at no expense of the other. Our stretch goal was to further improve the accuracy and usefulness of medical reasoning feedback. After extensive testing, we selected the Google Gemini API as our medical reasoning judge. As a natural extension of this work, we plan to train a specialized medical reasoning discriminator to provide more precise and controllable feedback in future iterations.

III. LITERATURE REVIEW

The paper *QLoRA: Efficient Finetuning of Quantized LLMs* [1] introduces the design motivation and implementation principle of QLoRA, which solves the challenges we faced in our project: the limitation of using only a single 24GB GPU for inference and training per task submission. This made it difficult to use full-parameter fine-tuning, so we adopted QLoRA to train both the discriminator and generator. For the selected base model, *Meta-Llama-3.1-8B-Instruct*, this approach almost completely kept training quality while significantly reducing computational requirements, allowing us to train the model effectively and achieve project goals even with only a 24GB single GPU.

In another paper, *SELF-REFINE: Iterative Refinement with Self-Feedback* [3], shows a idea of using a single LLM model to self-evaluate generated content and then modify it based on the evaluation, repeating the process until a modification limit is reached or no further modifications are needed. Our project architecture is based on this idea. We use two discriminators to provide feedback on realism and medical reasoning for the initially generated notes. If no problems are found, the feedback is `good`. We then combine that feedback into the generator’s prompts, allowing the generator to update the initial note based on the feedback, fixing medical mistakes and outputting the final optimized note. This significantly improves the quality of the medical notes generated. However, unlike the original paper [3], due to the complexity of medical logic and the performance limitations of the base model, we were unable to use a single LLM model to accomplish everything. Instead, separately fine-tuning a discriminator to evaluate note quality allows us to better identify medical mistakes in a note.

IV. DATASET

A. Preprocessing and Data Construction

We use a publicly available MIMIC-IV-2 hosted on Kaggle, which is a subset of the original MIMIC-IV dataset [2]. We first pre-process the dataset by extracting and parsing the discharge note portion of each record. The notes are normalized into a structured representation with consistent sections and fields. This pattern controlled modification and regeneration while preserving the original linguistic patterns of clinical documentation.

Since MIMIC-IV is fully de-identified, we augment the discharge notes by injecting synthetic protected health information (PHI) to restore realistic identifiers context. For example, Patient and clinician names are generated using in a random but consistent manner. For fields that require semantic coherence, including admission date, discharge date, date of birth, discharge instructions, and follow-up instructions, we use **LLaMA 3.1 8B** to generate context-aware values conditioned on the surrounding clinical text. This hybrid approach is aimed to produce discharge notes.

Based on this pipeline, we construct two dataset versions: a good one and a bad one. The good dataset consists of high-quality synthetic discharge notes with internally consistent PHI, formatting, clinical content. It is used to fine-tune the generator to learn realistic language use and document structure. The bad dataset is created by intentionally injecting systematic errors into the PHI and note structure, including future dates, inconsistent temporal relationships, mismatched patient or clinician names across sections, and degraded formatting that makes the document appear unstructured. Each bad example is paired with localized error annotations indicating where the inconsistency occurs and why it is incorrect.

Each dataset contains 1,200 discharge notes. We split both the good and bad datasets into 1,000 training examples, used for fine-tuning and discriminator training, and 200 test examples for evaluation.

B. Realism Scoring

To support both training and evaluation, we assign each synthetic discharge note a *realism score* using a unified scoring pipeline. The realism score is designed to capture how closely a synthetic note resembles a real clinical discharge summary and is computed as a weighted combination of three complementary components: linguistic realism, document structure realism, and clinical consistency realism.

Linguistic realism evaluates PHI-level using a set of rule-based checks, including name formatting, capitalization, absence of invalid characters, date parseability, and temporal validity constraints. Document structure realism measures whether a discharge note follows expected formatting conventions, including header order, required sections, and attending physician information. Clinical consistency realism evaluates semantic coherence between patient attributes and clinical content.

Specifically, we assess the relevance of the primary diagnosis with respect to the patient’s age and biological sex using

LLaMA 3.1 8B, which assigns relevance scores that are normalized and combined with greater emphasis on age–diagnosis consistency.

The final realism score is computed using a mixed aggregation strategy that combines a weighted geometric mean and a weighted arithmetic mean over the three components, ensuring that low scores in any single dimension significantly reduce the overall realism score while maintaining numerical stability.

Both the good dataset and the bad dataset are scored using the same pipeline. Importantly, all generator outputs are evaluated using the identical realism scoring procedure, ensuring fair and consistent comparisons across training data, discriminator supervision, and generated samples.

V. MODEL ARCHITECTURE

Our system contains a Generator that generate discharge medical notes containing PHI, and a hybrid Discriminator framework designed to evaluate the realism and medical consistency of the generated notes. The Generator and the Local Discriminator are built by the Llama-3.1-8B-Instruct base model and are fine-tuned using parameter-efficient Quantized Low-Rank Adaptation (QLoRA). Complementing the local model, we incorporate a second External Discriminator is the Gemini API, which serves as a high-level model to detect medical logic.

A. Generator Architecture

we use Llama-3.1-8B-Instruct as our base model. After fine-tuning on labelled data, it can generate unstructured discharge summaries with Personal Health Information (PHI) by receiving instructions.

B. Hybrid Discriminator System

To maximum our reasoning with limited resources, we use a dual-discriminator approach:

- 1) **Local Discriminator (Llama-3.1-8B-Instruct):** We fine-tune Llama-3.1-8B-Instruct to be a realism discriminator. This model can determine the realism of the generated medical notes and produce a structured JSON about the reason of failed notes.
- 2) **External Discriminator (Gemini):** For complex medical logic, we use the Gemini Pro API. We compared its performance against our fine-tuned model and found that Gemini was significantly better at handling reasoning tasks. And We chose Gemini as the final model to check for medical logic.

C. Parameter-Efficient Adaptation (QLoRA)

We use QLoRA to fine-tune our Model (the Generator and Discriminator) on OSCAR. This method freezes the base model weights in 4-bit NormalFloat (NF4) precision and injects trainable LoRA adapters.

And We apply these adapters to *all* linear projections in the every transformer layers:

$$\Theta_{\text{target}} = \{q, k, v, o, \text{gate}, \text{up}, \text{down}\}_{\text{proj}}.$$

For a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the forward pass is approximated as:

$$W'x = (W_{\text{frozen}} + \Delta W)x = W_{\text{frozen}}x + \frac{\alpha}{r}BAx, \quad (1)$$

where B and A are low-rank trainable matrices.

D. Training Objective (SFT)

We fine-tune the model using standard Supervised Fine-Tuning (SFT). The training data consists of instruction-output pairs, which are the official Llama-3 chat template.

VI. TRAINING PROCEDURE

A. Objective

We trained both models on a supervised next-token prediction task.

In the case of the Generator, we would like to have a model that can comprehend everything from patient JSON profile all the way down to discharge summary. Thus, we laid down the losses over the entire sequence.

With the Discriminator, we use completion-only supervision. We mask out the input prompts as they are long and with some repetition, here we only compute the loss on the assistant output (the JSON critique). This makes the model completely focused on its realism.

B. Optimization

We used the Paged AdamW optimizer (32-bit), which helps manage memory efficiently. We employed a cosine learning rate schedule to smooth out the training process. The learning rate starts at 2×10^{-4} after a warmup period covering 3% of the total training steps.

C. Batch Size and Epochs

Both models were trained for 3 epochs on a single GPU. To fit the models into memory while maintaining training stability, we used a small per-device batch size combined with gradient accumulation:

- **Generator:** Batch size of 2 with 4 accumulation steps, resulting in an effective batch size of 8.
- **Discriminator:** Batch size of 2 with 8 accumulation steps, resulting in an effective batch size of 16.

D. Regularization

We also regularized the model to avoid overfitting (a challenge we faced as our dataset was quite small):

- **LoRA Dropout:** We used 0.1 for the Generator dropout to promote creativity, but 0.05 for the Discriminator in order to achieve more stable outputs.
- **Weight Decay:** Common value of 0.001.
- **Gradient Clipping:** Limited to 0.3 to avoid exploding gradients.
- **Efficiency Tricks:** We also employed sequence-length grouping and gradient check pointing to accelerate training and reduce the memory consumption.

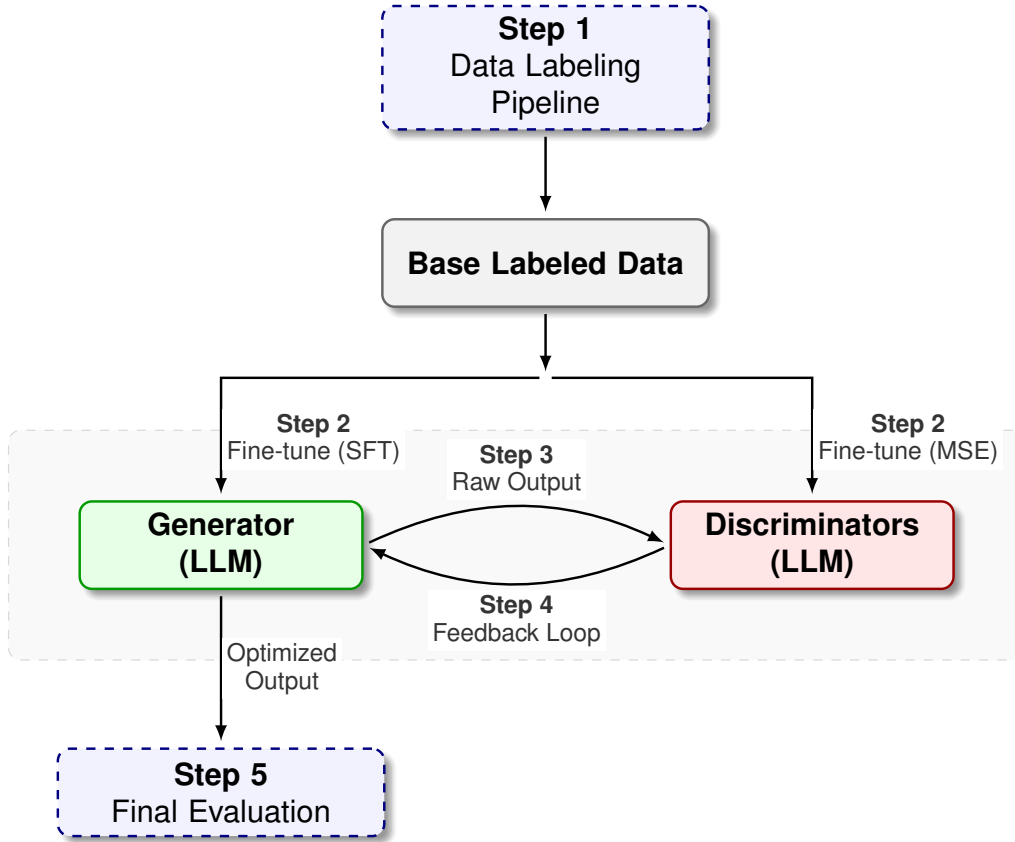


Fig. 1: System Workflow Architecture.

E. Evaluation Metrics

We evaluate our system using a mix of automated metrics and model-based judging. Initially, we assessed the realism of the generated notes using the original annotation pipeline to maintain a controlled variable for comparison against the baseline system. For the more challenging aspect of medical logic, we utilized the Gemini API to evaluate the results, serving the same role as a discriminator.

VII. RESULTS

Overall, these results have indicated the effectiveness of discriminator-guided generation over prompt-only method.

A. Quantitative Results

Table I summarizes the performance of all models and reveals some interesting patterns. Firstly, the baseline LLaMA-3.1 model with a standard one-shot user prompt gets a steady high realism of 0.9146. Although it captures discharge summary formatting and PHI structure well, it struggles with medical reasoning (0.217). Adding medical constraints to the generator prompt improves medical reasoning substantially (0.503), but causes realism to drop (0.5796). It is clear that the notes become more logically correct but lose some of their natural clinical style.

Our fine-tuned generator without a discriminator has seen improvement in both metrics, with a realism of 0.8879 and

medical reasoning of 0.446. Finally, we can see that the generator with dual-discriminator (our "Gemedi" model) performs best overall, having the highest realism (0.9252) and medical reasoning (0.526) scores. Discriminator feedback appears to greatly enhance clinical logic while keeping notes realistic.

TABLE I: Model Performance Comparison

Model	Realism Score	Medical Reasoning Score
Baseline	0.9146	0.217
Baseline + Gen Prompt	0.5796	0.503
Gen w/o Discriminator	0.8879	0.446
Gen w/ Discriminator	0.9252	0.526

B. Qualitative Analysis

As shown in Figure 2 and Figure 3, we see that baseline models demonstrate a noticeable tradeoff between medical reasoning and realism. Even though adding medical constraints improves medical reasoning, they often result in unnatural medical notes or irregular formats. On the contrary, we can see our generators are doing better. The fine-tuned generator without discriminator but with optimized prompt has already seen a rise in both realism and medical reasoning. In addition, we're able to see the discriminator-guided generator is able to keep discharge note realism while being better at correcting logic and temporal hallucinations.

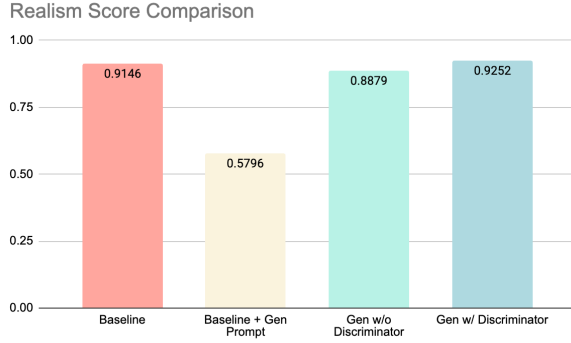


Fig. 2: Realism score comparison showing generator with discriminator outperforms baseline models across all realism dimensions.

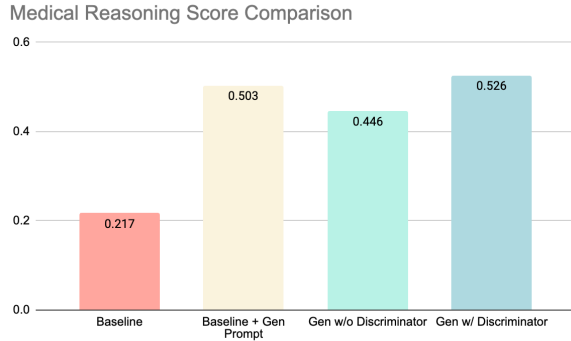


Fig. 3: Medical reasoning score comparison showing generator with discriminator achieves comparable-to-superior scores than baseline models.

C. Goal Achievement

We successfully met our base and target goals. As shown in Table I, our finetuned generator without discriminator already outperforms the baseline in both dimensions, confirming we didn’t degrade performance. More importantly, our discriminator-guided generator achieves the highest realism of 0.9252 and medical reasoning of 0.526 scores simultaneously, surpassing all baseline approaches and fulfilling our target goal of balancing both metrics.

For our stretch goal, we made partial progress. After extensive testing with different judge models, we selected the Google Gemini API as our medical reasoning discriminator based on its ability to provide detailed, clinically-grounded feedback. However, training our own specialized medical reasoning discriminator would provide more precise, controllable feedback, which is planned for future work.

VIII. CHALLENGES

During the project, we encountered two main challenges. The first was resource and memory bottlenecks. In Oscar, student accounts only allowed the use of one 24GB GPU per task, which made it difficult to perform full-parameter fine-tuning of the model. Moreover, we could not use larger,

smarter models, such as Llama-3.1 (70B), as the base model for our project.

To address this challenge, we first selected an LLM (Meta-Llama-3.1-8B-Instruct) that performed best in medical scenarios under the 24GB VRAM constraint. During model training, we used QLoRA technology to significantly reduce the cost of model fine-tuning while ensuring that the fine-tuning performance was almost as good as full-parameter fine-tuning.

The second challenge was the performance limitations of the discriminator. After training the discriminator using QLoRA, we found that the model performed well in detecting surface-level linguistic and formatting issues, but it failed to accurately detect some complex medical logical problems, such as a diagnosis of “Hypertension” followed by a prescription for “Insulin.” This was a very serious problem because even if all PHI in the generated medical note was logically consistent, the content itself did not conform to medical logic, resulting in the whole note being useless in real-world applications.

This problem occurred because we lacked sufficient medical notes labeled with complex medical logical errors to train the discriminator. Additionally, Meta-Llama-3.1-8B-Instruct did not support providing accurate and in-depth logical feedback on medical texts as a base model. Based on this situation, we decided to use a powerful model like Google Gemini to comprehensively evaluate the medical reasoning of the discharge notes generated by the generator and provide detailed feedback. This feedback was used as part of the prompt for refining the generator, resulting in final discharge notes that performed very well in both PHI consistency and medical reasoning.

IX. ETHICS

This project is closely relative to ethical issues surrounding privacy concerns, as mentioned in the *motivation* section of this paper. To protect patient privacy, the medical field lacks complete and high-quality medical note data, which leads to insufficient training data for medical-related artificial intelligence, hindering the development of medical AI. This project effectively addresses this issue because it avoids the direct use of real patient-identifiable data by design. At the same time, it continuously generates new training data containing virtual patient information, significantly mitigating the contradiction in medical AI research where “data are most valuable but most inaccessible.”

However, there are some potential ethical issues with this project that we need to consider. Firstly, this generator cannot guarantee the generation of medical notes that fully conform to realistic medical logic. This means that the generated data may contain some difficult-to-detect medical errors. If this data is directly used to train other medical-related models and applied in real-world settings, these errors could pose risks to patient safety.

Secondly, all training data come from a single dataset, which may lead to the amplification of potential biases in the training data during training, resulting in biased medical note

generation. Therefore, based on the current training results and ethical risks, our generated medical notes can only be used as a supportive research tool and are not yet capable of participating in real clinical treatment processes.

X. REFLECTION

Overall, we believe that the project was successful in achieving its research objectives, but it still has clear limitations in medical reasoning. The final generated notes achieved consistently high realism scores, showing that the current generator can produce discharge notes with correct formatting and plausible PHI values. Moreover, the strong performance of the discriminator on the test set demonstrates that the QLoRA-based discriminator is stable in detecting surface-level errors. Comparative results further show that the Self-Refinement architecture combined with external evaluation from Gemini significantly improves the overall quality of the generated medical notes. However, due to limitations of the base model, medical reasoning scores never reached a satisfactory level. As a result, among the three goals defined in the introduction section, we successfully achieved the first two: the base goal and the target goal.

Initially, we aimed to use a single discriminator to provide feedback on both realism and medical reasoning, but the results were not good. We also planned to use a reinforcement learning architecture with a discriminator and generator, but due to practical training limitations, we found that refinement was a more feasible approach.

For future work, we aim to improve in four main directions. Firstly, instead of relying solely on prompt-based refinement, we aim to directly use discriminator feedback to train the generator, allowing the model itself to improve rather than only refining its outputs. Secondly, we plan to train a local model to replace the Google Gemini API, which would eliminate API costs and significantly improve inference speed. Thirdly, with access to more powerful GPUs, we intend to fine-tune larger models, such as LLaMA-3 70B, to achieve substantially stronger medical reasoning performance than our current 8B model. Finally, to validate the practical value of our synthetic data, we plan to train external NLP models using only our generated discharge summaries. If these models perform well on real-world datasets, it would demonstrate that synthetic data can effectively replace sensitive real clinical data.

Through this project, our most significant takeaway was learning how to build and evaluate a model pipeline from scratch. This included preprocessing raw data, selecting an appropriate base model, training models, and systematically evaluating and comparing results to assess model performance. Throughout the whole process, we learned many useful skills, such as preprocessing data, fine-tuning models, and prompt engineering, all of which were valuable takeaways.

XI. CONCLUSION

A. Main Contributions

We proposed a that is the Dual-Discriminator and Generator Feedback Loop. Through using different evaluators for the

structural realism and the medical reasoning, our refinement framework effectively product quality discharge notes which are linguistically fluent as well as clinically reliable.

B. Key Findings

Base models perform well in realism but poorly on medical reasoning. While applying prompt constraints improves reasoning, it causes a significant drop in realism. And Our system achieves substantial improvements in both realism and medical logic.

C. Implications

This task relates to the lack of privacy-preserving medical data. We hope that by generating believable synthetic discharge notes, we can train downstream clinical AI models without the legal and ethical overheads of working with real patient records (PHI).

D. Final Thoughts

In the future, we plan to reduce API dependence by training custom local reasoning models and investigate larger architectures (e.g., Llama-3 with 70B parameters) for more powerful native performance at a lower cost.

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